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AvexNet

Architecture

Installing the AvexNet System Platform

White Paper A47.01

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Introduction

This guide is intended to help the initial developer install the required platform to develop Java Web Applications using AvexNet. Any set up procedures related to installing and building Foundation as its own project can be found in the Foundation Installation Guide.

System Requirements

It is expected that you have the minimum system configuration to develop applications using AvexNet:

- Intel® Celeron® D Processor 320 (2.40GHz, 533 FSB)
- Microsoft® Windows® XP Professional Edition
- 1GBB DDR SDRAM at 400MHz
- Internet Explorer 6.0
- Visual Source Safe Client 6.0a

Software Development Platform

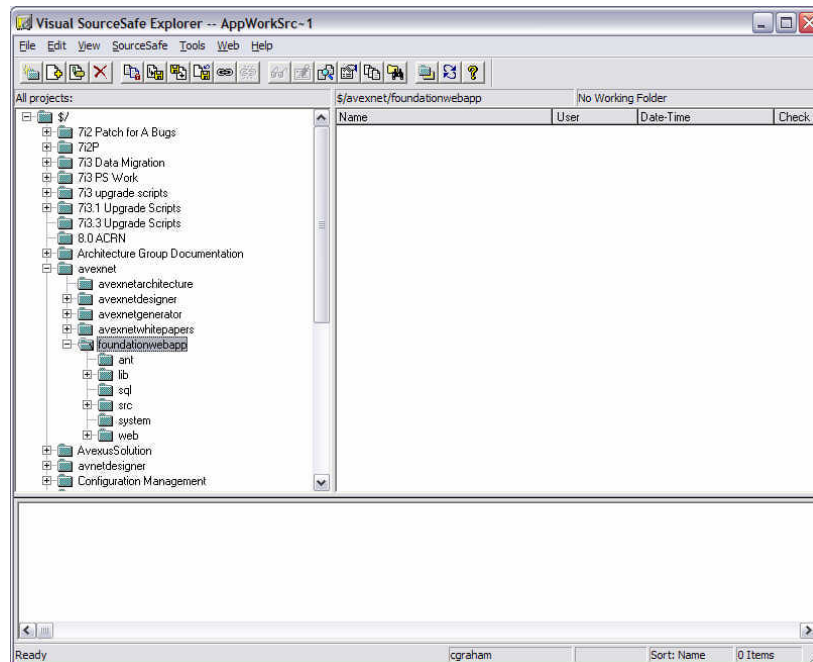
The following software development platform is recommended for AvexNet:

- Java 2 SDK 1.4.2
- Eclipse 3.0
- JBOSS 3.2.5 (Tomcat 5)
- Visual Source Safe Client 6.0a

Installing System Software

A distribution of all recommended system software is available in SourceSafe as part of Foundation. In order to obtain these distributions the developer will need to use Visual SourceSafe to obtain the foundation project first. If Visual Source Safe Client 6.0a has not been pre configured on your system please see an IT administrator immediately.

Step 1 – Open Visual Source Safe Client 6.0a

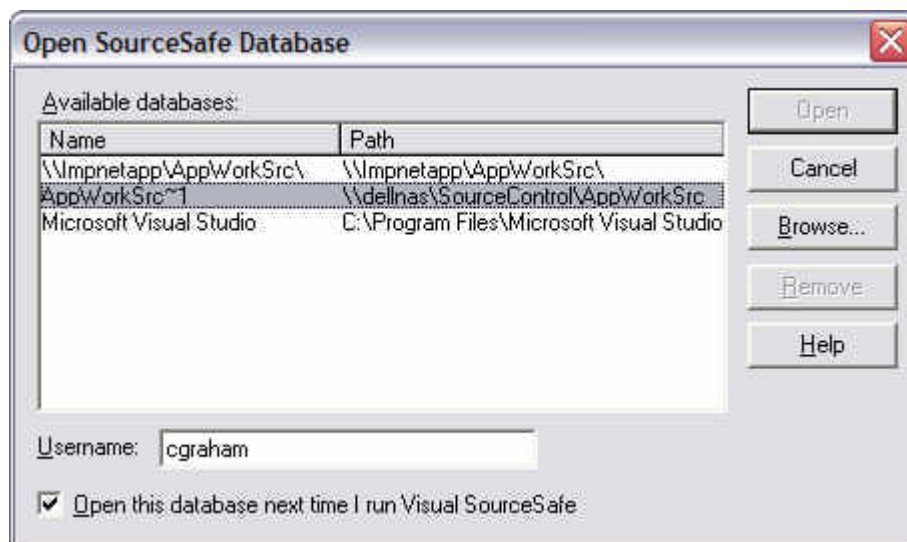


You will need to use the following login info for VSS to use the repository:

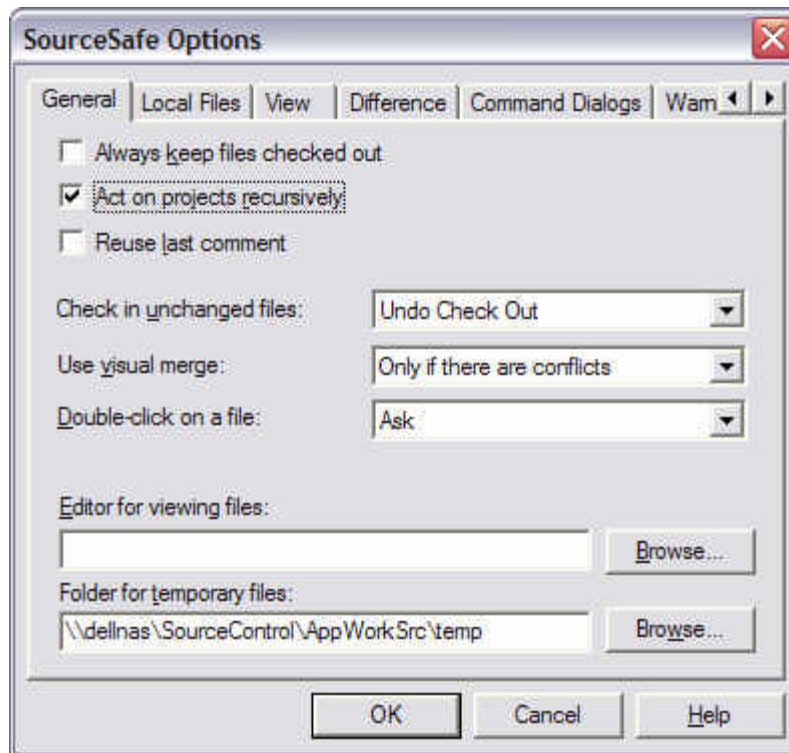
Server: \\dellnas\SourceControl\AppWorkSrc

User: <ntusername>

Password: <ntpassword>

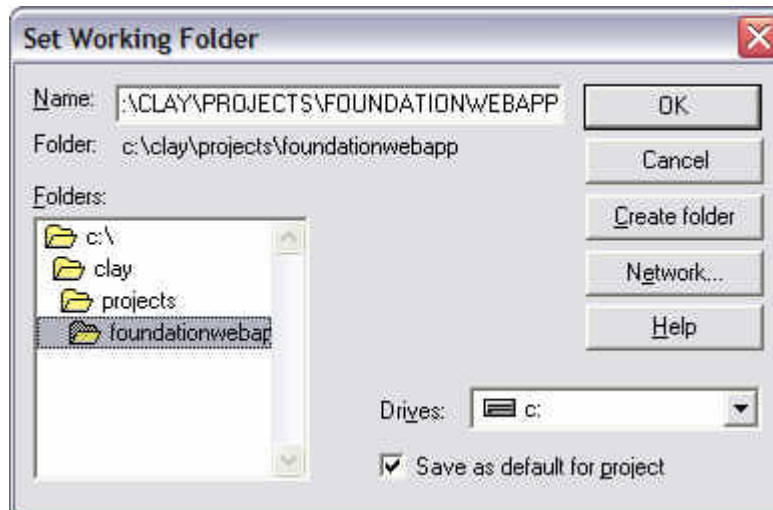


Use the following project options prior to getting the project:



Once you have opened the VSS client you will need to map the \$/avexnet/foundationwebapp directory to your local directory. Commonly this is mapped to a local projects directory where you will keep all your Eclipse projects. ***It is recommended that you create the following directory for your project:***

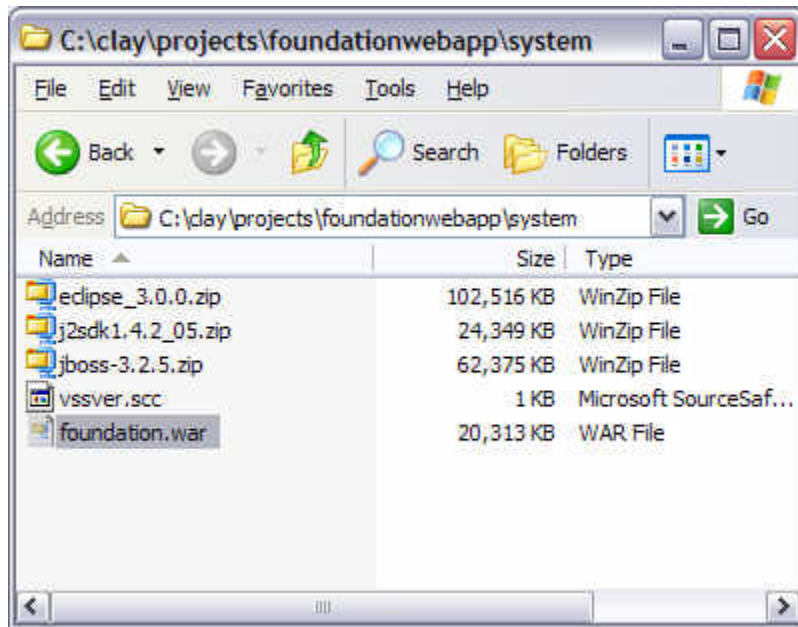
C:\<ntusername>\projects\foundationwebapp



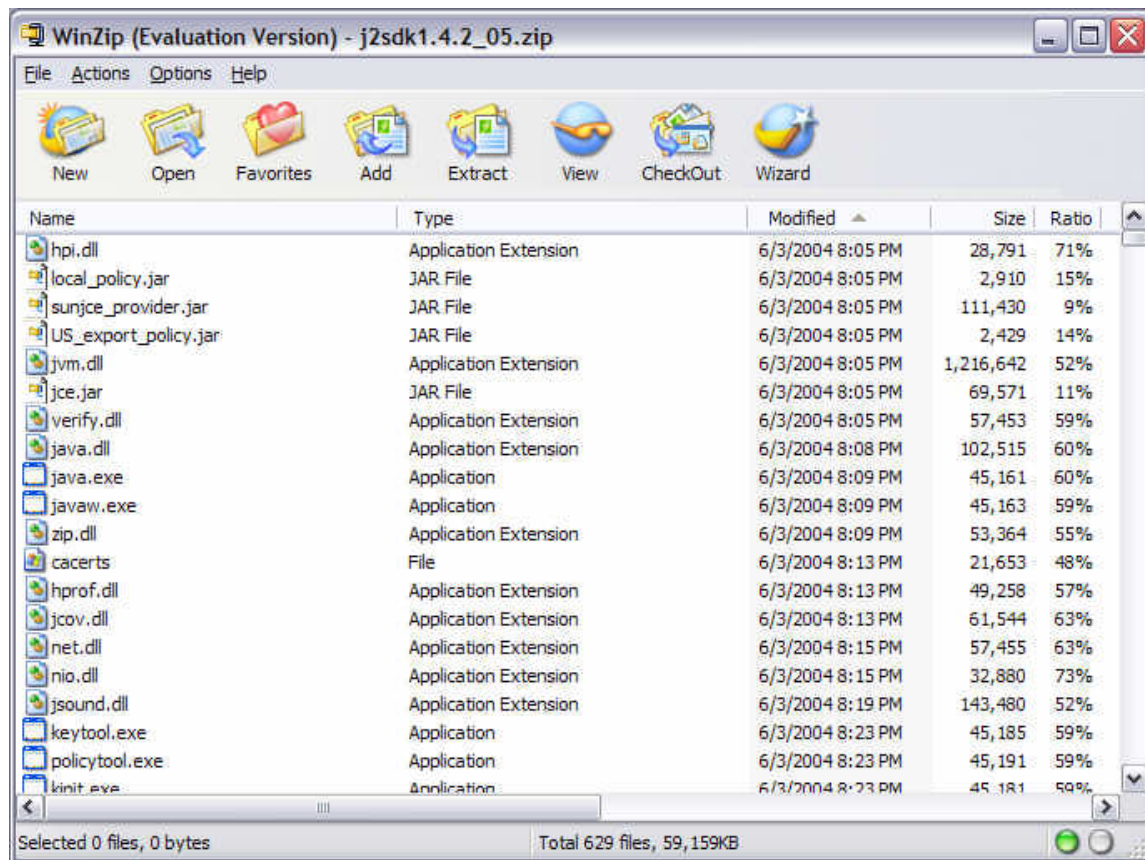
Once you have mapped to your local directory you can run Get Latest Version... by selecting the foundationwebapp directory in VSS and right clicking, a menu item will pop up that allows you to get the latest version of Foundation.

Step 2 – Install Java JDK 1.4.2

Once you have gotten all of foundation you will be able to install the Java JDK. To begin go to the system directory in foundation:

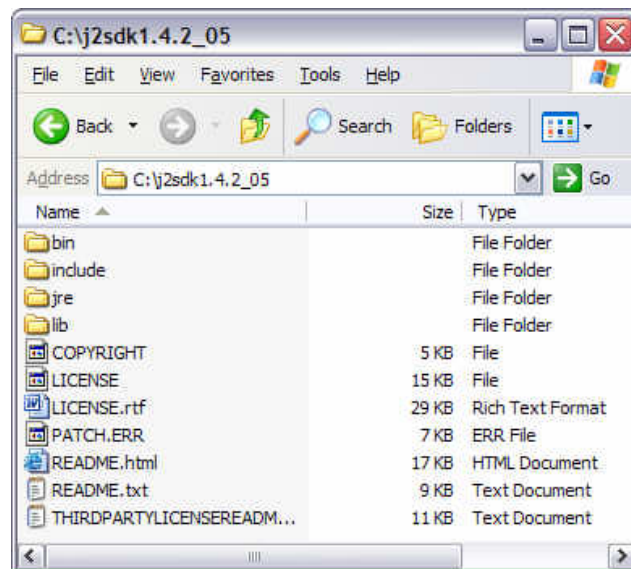


Open the ZIP archive j2sdk1.4.2_05.zip

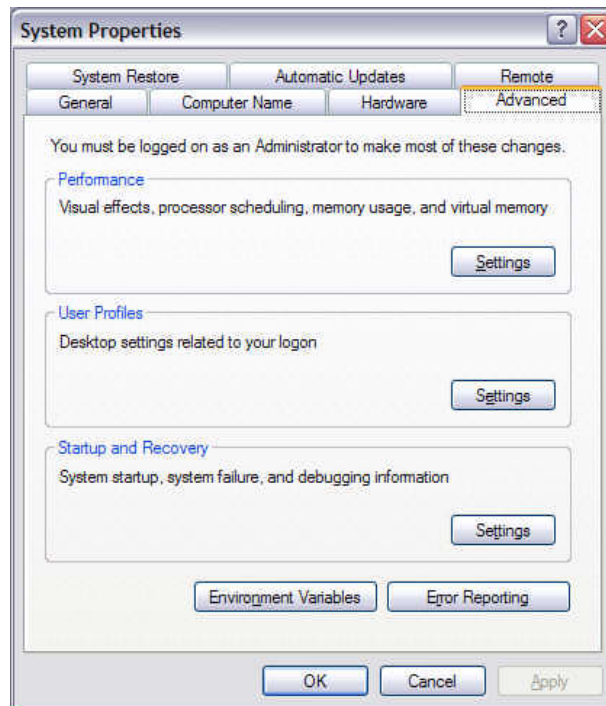


Next extract the JDK to the directory you would like it to be installed. . It is recommended that you install it to the C:\ directory so it will create the directory:

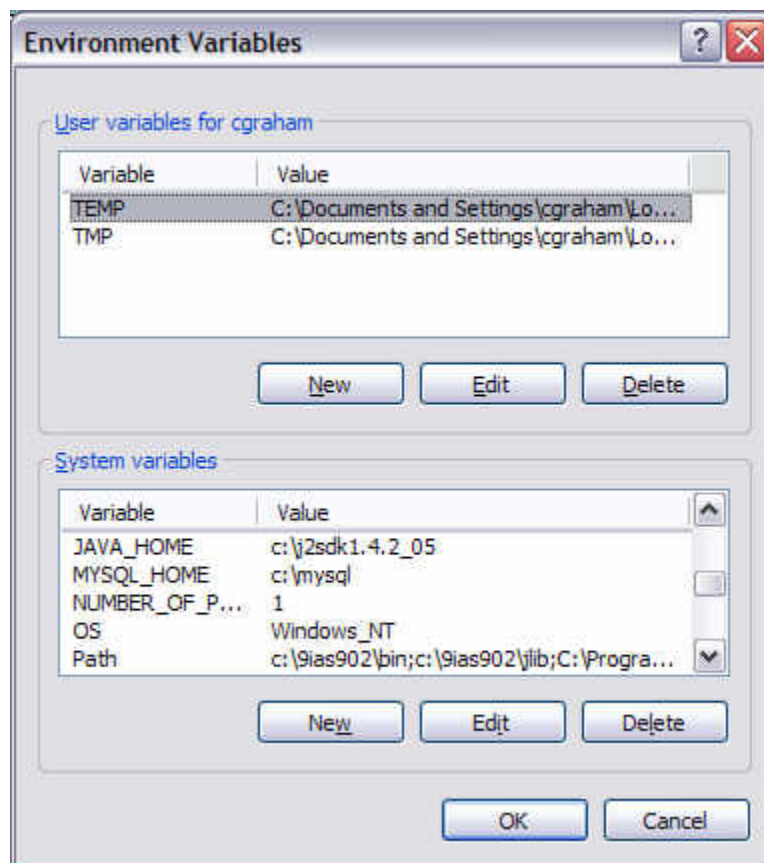
C:\j2sdk1.4.2_05



Control Panel>System Properties



Finally set the environment variable for `JAVA_HOME = C:\j2sdk1.4.2_05`



Step 3 – Install and Start JBOSS

Introduction to JBOSS

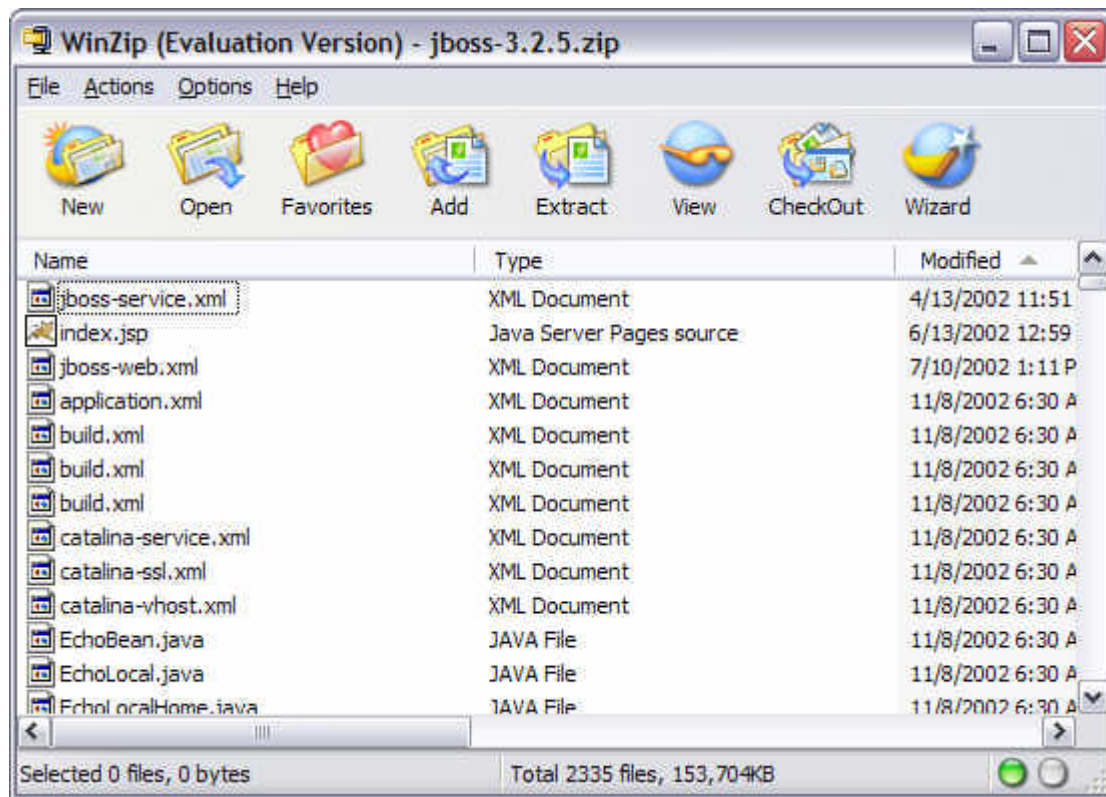
JBoss Application Server has evolved into a full feature and robust J2EE application server that is no longer limited to just the development lab anymore as evidenced by a dramatic increase in large-scale production deployments. A recent SDTimes survey showed that JBoss has expanded its presence in the production market from 15% in 2002 to 25% in 2003, earning it not only the third position in the application server market but also the honor of fastest growing application server in the industry.

The momentum behind JBoss can be attributed to a number of factors including:

- Open source, no cost product licenses. JBoss is free to download and free to use regardless of the size of your deployment.
- Scalability. Companies like National Leisure Group chose JBoss and JBoss Inc. support to help their applications scale both technically and economically. JBoss offers full clustering of any Java object (EJB, HTTP, POJO)
- Performance. Based on their own internal benchmark tests, companies like La Quinta Corporation have found that JBoss offers improved performance and superior server utilization to other leading J2EE application servers.
- Modular and elegant architecture. JBoss offers a unique architecture built on a microkernel-based design that leverages JMX extensions. The result is a lightweight component model that delivers advanced class-loading features and full lifecycle management.
- MBean Services-Oriented Architecture. Services can be easily added or removed based on your specific needs. All services are neatly packaged and fully hot-deployable. You can also create and add your own services easily.
- Aspect-Oriented Programming (AOP) Model. JBoss has a long running tradition of utilizing interceptors in the architecture that further add to the flexibility of the architecture.
- Additional application-level features such as No Compilation which are extremely popular among advanced Java developers. Also included is a full security implementation and JAAS integration.
- Standards support. JBoss Application Server offers full support for J2EE 1.4.

Installing JBOSS

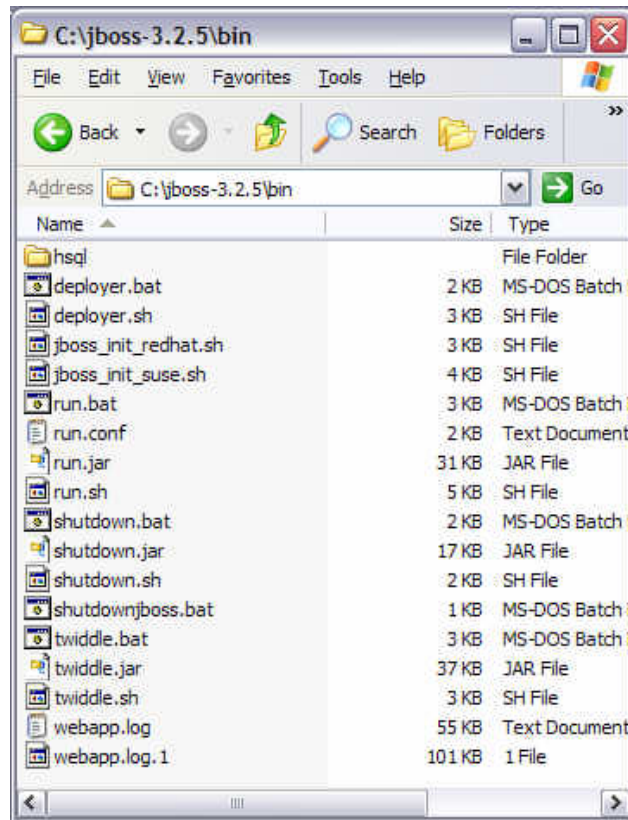
Open the ZIP archive `jboss-3.2.5.zip`



Next extract the JBoss application server to the directory you would like it to be installed. . It is recommended that you install it to the C:\ directory so it will create the directory:

C:\jboss-3.2.5

You will then want to start JBOSS. You can find the startup script in:



The following scripts control the startup and shutdown of JBOSS

- **run.bat** – The startup script. Runs the application server on port 8080
- **shdownjboss.bat** – Shuts down any JBOSS processes.

Finally Startup the server:

```
jboss start
10:38:58,233 INFO [SQLExceptionCodesFactory] Driver Version is 1.7.2
10:38:58,363 INFO [DefaultListableBeanFactory] Creating shared instance of singleton bean 'OrganizationTreeBLSERVICE'
10:38:58,363 INFO [DefaultListableBeanFactory] Creating shared instance of singleton bean 'OrganizationTreeBLTarget'
10:38:58,383 INFO [DefaultListableBeanFactory] Creating shared instance of singleton bean 'UiBLSERVICE'
10:38:58,383 INFO [DefaultListableBeanFactory] Creating shared instance of singleton bean 'UiBLTarget'
10:38:58,393 INFO [DefaultListableBeanFactory] Creating shared instance of singleton bean 'poplistDao'
10:38:58,403 INFO [SQLExceptionCodesFactory] Looking up default SQLExceptionCodes for DataSource
10:38:58,413 INFO [SQLExceptionCodesFactory] Database product name found in cache for DataSource [org.apache.commons.dbcp.BasicDataSource@9b774e1. Name is 'HSQL Database Engine'.
10:38:58,413 INFO [ContextLoader] Using context class [org.springframework.web.context.support.XmlWebApplicationContext] for root WebApplicationContext
10:38:58,424 INFO [ContextLoader] Published root WebApplicationContext [org.springframework.web.context.support.XmlWebApplicationContext: displayName=[Root XmlWebApplicationContext]; startup date=[Tue Nov 02 10:38:55 PST 2004]; root of Application Context hierarchy; config locations=[/WEB-INF/webContext-hibernate.xml]; ] as ServletContext attribute with name [interface org.springframework.web.context.WebApplicationContext.ROOT]
```

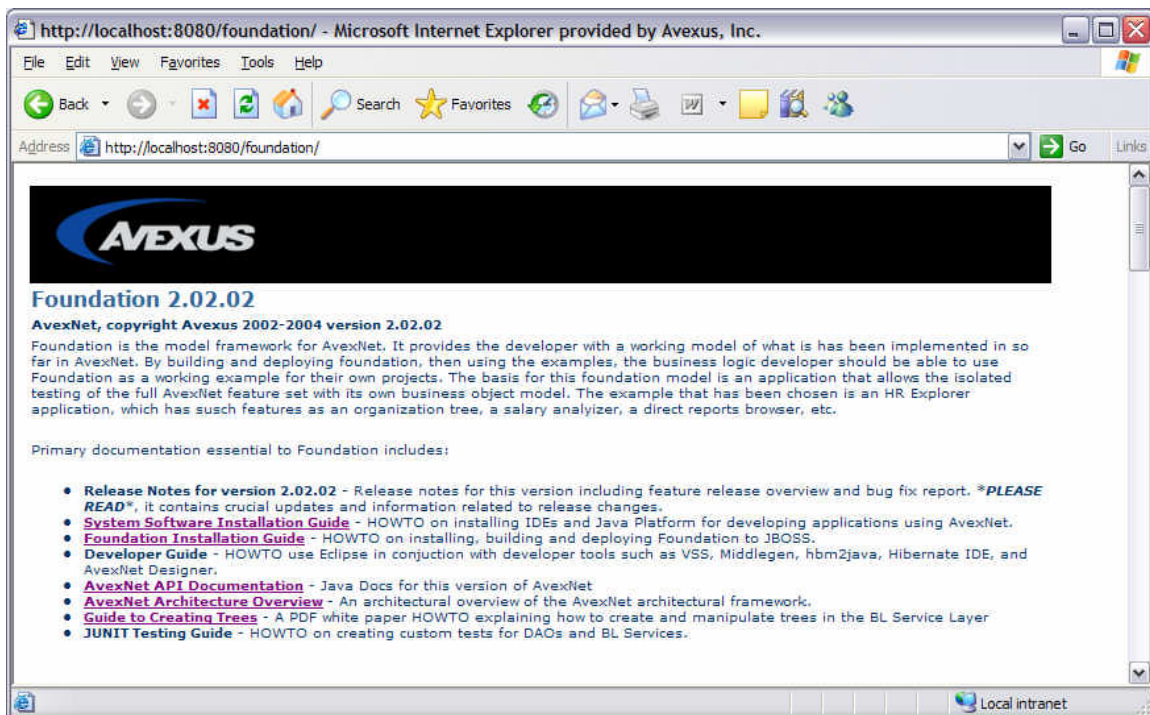
Running Foundation for the First Time

To run Foundation for the first time you will need to copy it to the deploy directory of JBOSS, this can be found at: `$\foundationwebapp\system\foundation.war`

Copy this file to: `C:\jboss-3.2.5\server\default\deploy` from that point wait a few minutes to let the application deploy.

You can test that the application server has started by going to the foundation home page at:

<http://localhost:8080/foundation/>



From this point it is advised that the developer continue on through the web application documentation to the online guide: **[Foundation Installation Guide](#)**.

Step 4 – Installing Eclipse 3.0

Introduction to Eclipse

Eclipse is an open platform for tool integration built by an open community of tool providers. Operating under a open source paradigm, with a common public license that provides royalty free source code and world wide redistribution rights, the eclipse platform provides tool developers with ultimate flexibility and control over their software technology.

Eclipse has formed an independent open eco-system around royalty-free technology and a universal platform for tools integration. Eclipse based tools give developers freedom of choice in a multi-language, multi-platform, multi-vendor environment. Eclipse provides a plug-in based framework that makes it easier to create, integrate and utilize software tools, saving time and money. By collaborating and exploiting core integration technology, tool producers can leverage platform reuse and concentrate on core competencies to create new development technology. The Eclipse Platform is written in the Java language and comes with extensive plug-in construction toolkits and examples. It has already been deployed on a range of development workstations including Linux, QNX, OSx and Windows based systems.

Although the Eclipse Platform has a lot of built-in functionality, most of that functionality is very generic. It takes additional tools to extend the Platform to work with new content types, to do new things with existing content types, and to focus the generic functionality on something specific.

The Eclipse Platform is built on a mechanism for discovering, integrating, and running modules called *plug-ins*. A tool provider writes a tool as a separate plug-in that operates on files in the workspace and surfaces its tool-specific UI in the workbench. When the Platform is launched, the user is presented with an integrated development environment (IDE) composed of the set of available plug-ins.

The quality of the user experience depends significantly on how well the tools integrate with the Platform and how well the various tools work with each other.

The Eclipse Platform (or simply "the Platform" when there is no risk of confusion) is designed and built to meet the following requirements:

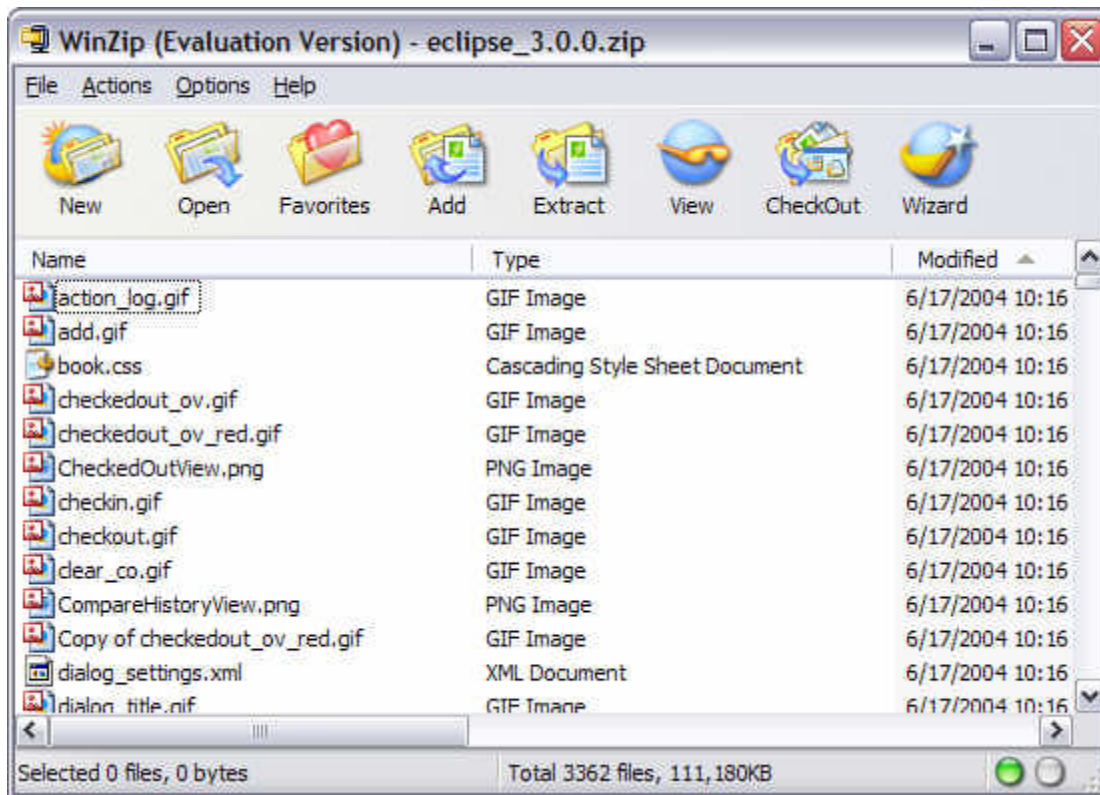
- Support the construction of a variety of tools for application development.
- Support an unrestricted set of tool providers, including independent software vendors (ISVs).
- Support tools to manipulate arbitrary content types (e.g., HTML, Java, C, JSP, EJB, XML, and GIF).
- Facilitate seamless integration of tools within and across different content types and tool providers.
- Support both GUI and non-GUI-based application development environments.
- Run on a wide range of operating systems, including Windows® and Linux.
- Capitalize on the popularity of the Java programming language for writing tools.

The Eclipse Platform's principal role is to provide tool providers with mechanisms to use, and rules to follow, that lead to seamlessly-integrated tools. These mechanisms are exposed via well-defined API interfaces, classes, and methods. The Platform also provides useful building blocks and frameworks that facilitate developing new tools.

Eclipse Installation

Eclipse is a web development IDE that will be required for developing applications using AvexNet.

To begin installation open the ZIP archive `eclipse_3.0.0.zip`:



Next extract the Eclipse IDE to the directory you would like it to be installed. It is recommended that you install it to the C:\ directory so it will create the directory:

```
C:\eclipse
```

You will then need to edit the projects directory for Eclipse. This is the default place that Eclipse will expect all of your Java projects to reside. Its recommended that you use the same directory where you got the latest version of foundationwebapp as the projects dir so that you project dir will be:

```
C:\<ntusername>\projects
```

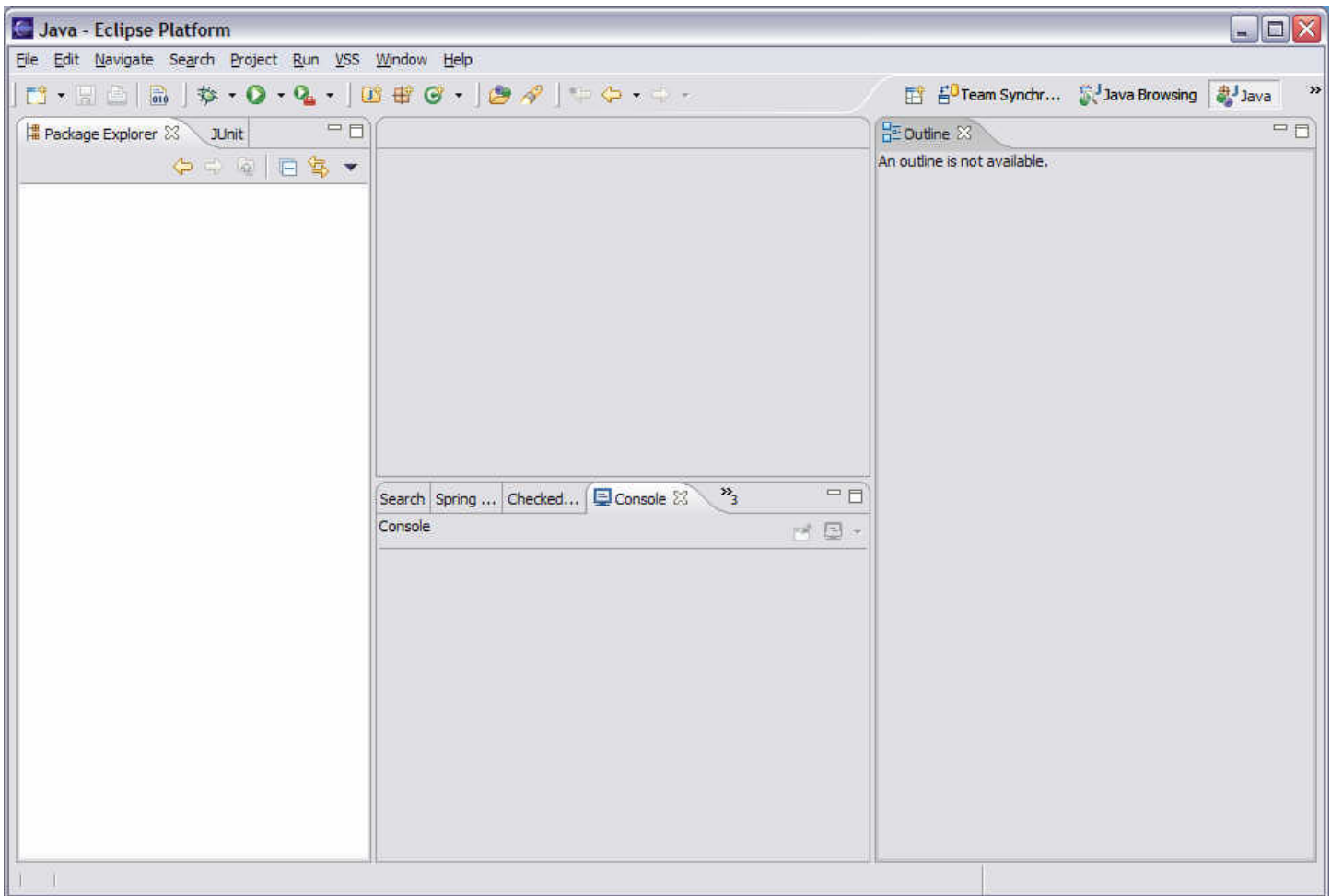
This is done by changing the project directory to use *your* Windows user name in the file:

```
C:\eclipse\configuration\org.eclipse.ui.ide\recentWorkspaces.xml
```

```
<?xml version="1.0" encoding="UTF-8"?>
<launchWorkspaceData>
  <protocol version="1"/>
  <alwaysAsk showDialog="0"/>
  <recentWorkspaces maxLength="5">
    <workspace path="C:\<ntusername>\projects"/>
  </recentWorkspaces>
</launchWorkspaceData>
```

Finally start Eclipse by running the batch script `C:\eclipse\run_eclipse.bat`

Eclipse should then start up normally and look like:



Conclusion

This concludes the HOWTO on installing system software, you should move onto the guide on installing and building Foundation from this point.